



Slovenian / University

Impact Report 2021–2026

Our first five years



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Director's foreword

A gauge exists to answer one question honestly: how much is left. This report covers the five years in which the School stopped assuming a printer's own reading did that, and started measuring the thing itself.



We chose an object almost too small to notice — the percentage on a shared office printer, glimpsed on the way past a photocopier alcove, trusted without a second look — because the smallest instruments are the ones an organisation checks least, and therefore the ones most likely to have drifted quietly out of true. Four fleet partnerships gave us an audit trail; the charts in the pages that follow set what a device actually does against what its gauge says it does.

The finding is not flattering. A reading calibrated once tends to stay wrong in the same direction, and a warning light asked to predict the future is, on the evidence collected here, mostly reporting the past. We record that plainly rather than soften it, because a School whose research subject is measurement owes its own report the same audit it performs on everyone else's dashboard.

Director, School of Continuous Improvement

By the numbers

Five years, and the four figures collaborators ask this School for most often.



340 publications

Peer-reviewed and practice output across the period, drawing on the School's measurement-systems and data-cultures themes.



\$18.4M in external grants

Funding secured across the fleet-and instrument-audit programmes this report describes, the five years to 2026.



112 partner organisations

Organisations that opened a device fleet, indicator, or gauge to the School's audit method since 2021.



27 international collaborations

Partner institutions engaged in comparative instrumentation work across the reporting period.

Numbers in context

Every figure in this report traces back to the same audit protocol: a sample of devices drawn from each partner's fleet, their displayed gauge reading logged, then a destructive toner-mass weighing taken within the hour to establish what was actually left. The protocol is deliberately blunt — weighing a cartridge tells the School nothing about **why** a gauge drifted, only **by how much** — and every commitment-shaped number in the case studies that follow was produced this way, not modelled or estimated. Where the four impact areas differ is in what they did once the gap was measured.

Two calibration checks ran across every partnership before a single figure was written up. First, an inter-rater check: two auditors weighed the same cartridge sample independently, and a reading stood only where the pair agreed within a fixed

tolerance. Second, a cross-site check: the same protocol, run at a partner's own facilities team's request, against a control cartridge of known mass shipped between sites. Neither check changed what the four case studies found; both are why the School is comfortable putting a number, rather than a description, on each finding.

About the School

The School of Continuous Improvement is the University's measurement-and-process school, tracing its lineage to the former Institute for Measurable Outcomes and gathering, under one roof, everyone who studies how an organisation counts and what the counting quietly does to it. Its six research themes — measurement systems and instruments, evaluation ecosystems, organisational learning and improvement, data cultures and dashboards, impact science, and anticipatory performance — share a single preoccupation: the point at which a number stops describing a thing and starts standing in for it.



Training runs through the Master of Applied Measurement, the School's flagship programme, and through Capability Sprints, its short-course line for practitioners who want the habits without the degree. Researchers here tend to turn the School's own instruments into research subjects as readily as a collaborator's, and this report's four case studies were chosen because they sit exactly on that line: ordinary

equipment, in ordinary offices, that nobody had thought to formally measure until somebody finally did.

The School's PhD candidates carried much of the fieldwork behind this report, each embedded with a partner's facilities or technology team for a full audit cycle rather than a single site visit. Several candidates report that the placement changed how they read every other indicator on a partner's premises, not only the one they were sent to check — an occupational habit the School regards as evidence the training is working as intended.

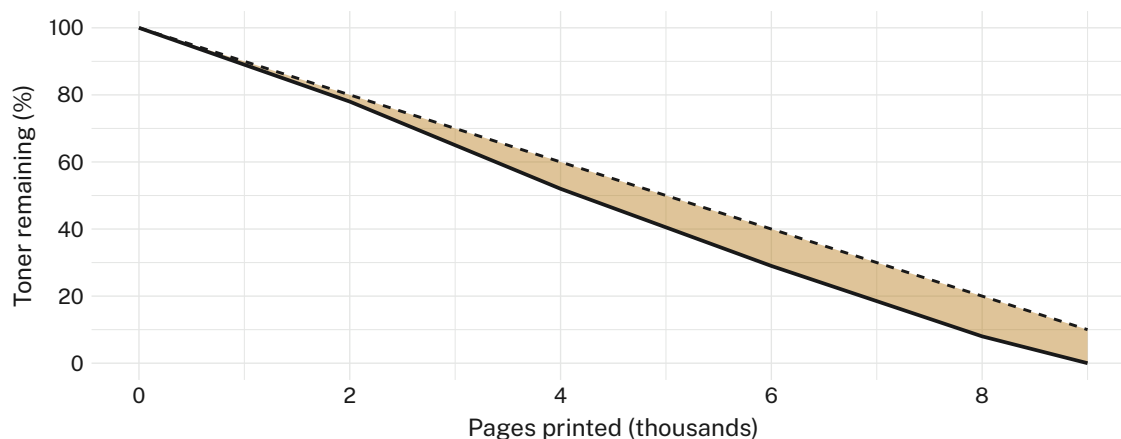
How we've delivered

The four areas below carry the case behind them; where a chart is available, it runs alongside the case it belongs to.

Reading the gauge

The Adaptive Metrics Lab's instrument-drift work usually studies indicators built for a purpose and left running past it. This period, Senior Lecturer Renke Sabel's team pointed the same method at a device most partner organisations had never thought to audit at all: the percentage gauge on a shared office printer, taken for granted precisely because nobody expected it to need checking.

Working with a big-four professional services firm's shared-services print fleet, the team weighed a sample of cartridges destructively against their displayed reading at four points across each cartridge's working life. The gauge's percentage tracked a fixed assumption about print density rather than the fleet's actual mixed load of dense spreadsheets and mostly-blank meeting agendas; the firm's helpdesk had logged the resulting mid-job failures as printer faults for three years before the audit reclassified them.



Displayed gauge reading against audited toner mass remaining, by pages printed since the cartridge was installed. The audit reaches zero a full two thousand pages before the gauge admits it.

When the light turns amber

The Review Cadence Observatory asks how often a standing review arrives too late to change anything; this period it turned that question toward a smaller cadence than a review cycle — the single warning a printer gives before it fails a job.



A federal infrastructure department asked the Observatory whether an early-warning ensemble could flag a fleet printer's failure before its own gauge did. Research Fellow Osei Vandermeer's team built a model from queue length, cartridge age, and

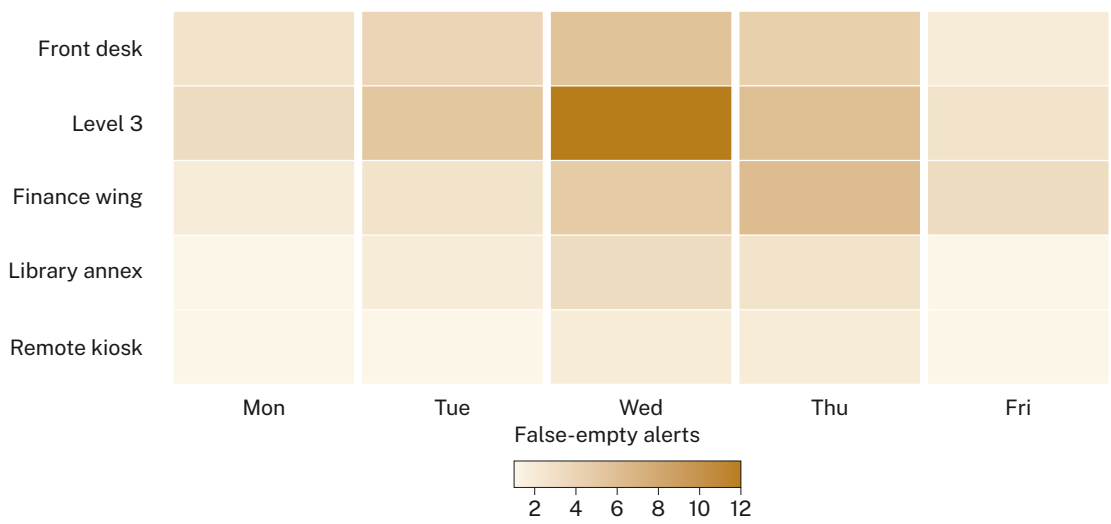
job size, and found it predicted an incomplete print job appreciably earlier than the gauge’s amber warning — though “appreciably earlier” turned out, in practice, to mean a little over one page.

“An amber warning light is not a forecast. It is a printer telling you, at the last possible moment, something it could have told you six pages ago.”
— Research Fellow Osei Vandermeer, Convenor, Review Cadence Observatory

The false-empty alert

The Living Dashboard’s researchers treat every tile on it, and every comparable board a partner installs, as an object of study rather than a finished product. This period’s fieldwork found one of the simplest tiles yet: a fleet’s own “replace cartridge” signal, wired straight into an uptime board with no audit in between.

A regional university network’s facilities team had connected its shared printers’ replace-cartridge signal directly to the Living Dashboard’s uptime tile, treating every alert as confirmed empty. Lecturer and Convenor Marit Osayande’s team logged each alert against an audited toner-mass reading taken within the hour, and found most midweek alerts fired while measurable capacity remained — the tile’s uptime figure was recording a queue of unnecessary cartridge swaps as unplanned downtime.

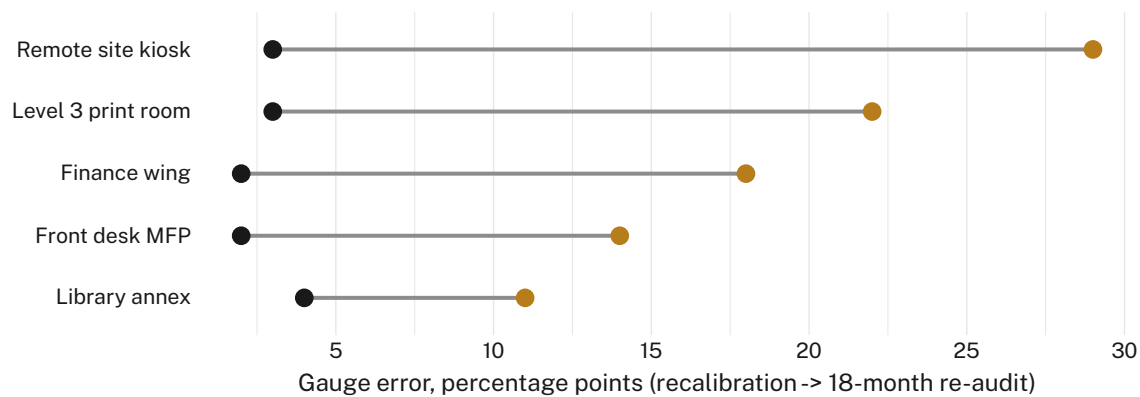


False-empty alerts logged per weekday across five fleet sites, one reporting quarter. The pattern tracks each site’s busiest print day, not its cartridges’ actual depletion.

Recalibration, then drift

Evaluation of Evaluation asks what a review actually changes, as distinct from what it merely re-certifies; applied to a device fleet, this period’s strand of that programme asked the same question of a gauge rather than a policy.

Working with a national telecommunications carrier’s back-office print fleet, Postdoctoral Fellow Joost Nwosu’s team recalibrated five device classes’ gauges against an audited baseline, then returned eighteen months later to re-audit the same devices. Every class had drifted back out of true; the remote-kiosk model drifted furthest, its gauge again reporting comfortably more toner than the audit could find.



Gauge error, in percentage points, at recalibration versus at the eighteen-month re-audit, across five device classes. Every class drifted; none drifted toward zero.

“Recalibration isn’t a fix, it’s a snapshot. Eighteen months later the same gauge had quietly gone back to telling the story it always told.”

— Postdoctoral Fellow Joost Nwosu

What’s next

Over the next planning period, the four partnerships in this report become the pilot cohort for a standing quarterly audit cycle rather than a one-off study: each partner’s fleet will carry a re-calibration date, after which a gauge’s continued reporting requires a fresh audit rather than an assumed carry-over. The Review Cadence Observatory will offer its early-warning ensemble as a fleet-wide service beyond the one department that piloted it, on the expectation that the lead time it buys stays measured in pages rather than minutes.



The Living Dashboard will pilot a rule decoupling an uptime tile from a raw alert count, so that a fleet’s “replace cartridge” signal must clear an audit threshold before it counts as downtime rather than a routine swap. Evaluation of Evaluation will extend the eighteen-month re-audit into a standing instrument, on the expectation — not yet tested — that a shorter interval between audits produces less drift, rather than merely catching the same drift sooner.

None of this promises a gauge that stops being wrong. Measurement, done honestly, tends to find more drift than any fleet gets around to correcting, and the School expects that imbalance to persist well past this reporting period. Its more modest aim is to make the gap between a reading and the thing it reads a little harder to miss.

With thanks

This report draws on the cooperation of partner organisations across government, industry, and the university sector who opened a working fleet or standing instrument to the School’s audit, several of them mid-way through a procurement cycle or system upgrade of their own. We thank the dozens of facilities and technology staff who located an original calibration record, arranged access to equipment mid-shift, or simply admitted that nobody currently owned the figure their organisation had been quoting.

We are grateful to the reviewers, internal and external, who read this report's case studies before either the readings or the recalibrations they describe reached a partner's own board table. And we thank the wider University community for its patience with a School that spent a portion of this reporting period pointing its own method at equipment nobody had asked it to look at.



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